2.10.4 Choosing the Best Branching Factor

For a mystery solved, there is still a big question standing: what is the best branching factor to speed up our Huffman encoding method? We digressed, delving into the details of the analysis of heaps, and somehow left the most important question behind.

To answer this question, there is only one way: looking at figure 2.26, showing the running time for the main method of our Huffman encoding library.

It shows three interesting facts:

1. The best choice seems to be `D==9`;
2. Choosing any value larger than 7 will likely be as good as;
3. While the max gain for `_frequency_table_to_heap` is 50%, and for `_heapify` even up to 80%, we merely get to a 5% here.

Notice how the chart for create_encoding looks more similar to method `_heap_to_tree` than to `_frequency_table_to_heap`: also considering point #3 above, the explanation is that the operation on the heap only contributes to a fraction of the running time, while the most demanding methods needs to be searched elsewhere (hint: for `create_frequency_table` running time depends on the length of the input file, while for the other methods it only depends on the size of the alphabet...).